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ASM1 Report

Analyzing the LA pseudoinstruction

LA instruction: la $a0, PURPLE_LABEL

1. $a0 = 0x1001004e ===>>> $a0 = 0x10010077
2. lui $1, 0x00001001 ===>>> ori $s4, $1, 0x00000077
3. 0x3c011001, 0x34240077

String Storage in Memory

String constant: PURPLE_LABEL = .asciiz “purple: “
Instruction: la $a0, PURPLE_LABEL

Ascii Codes in hex for first 4 characters in the string: 0x 70 75 72 70

String ended up at $a0. Had a value of 0x1001004e. Changed to a value of 0x10010078

Final word is 0x70727570, reverse order of original hex ascii code. They are found in the address 0x10010060 + Value (+18)
Investigation:

Something new I learn from this exercise is to comprehend the movement of values of addresses using registers. Quite fascinating to see the changes that go on within a few instructions. I believe it would prove to be useful in debugging code, looking where values went or hid away. I would assume this will come into use in regard to the prologue and epilogue snippets.